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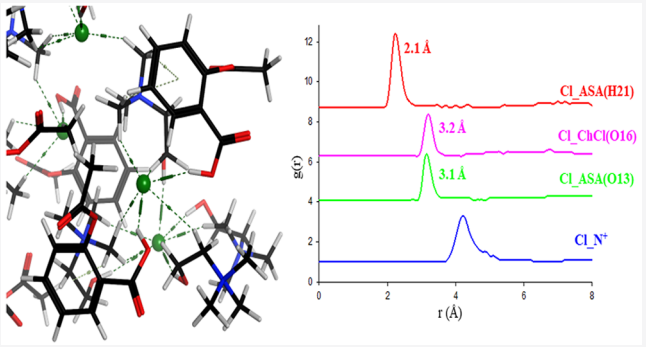
Molecular and Spectroscopic Insights of a Choline Chloride Based Therapeutic Deep Eutectic Solvent

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ABSTRACT: In this study, atomic level interactions of a 1:1 choline chloride (ChCl)/acetylsalicylic acid (ASA) therapeutic deep eutectic solvent (THDES) has been investigated by combining the molecular dynamics (MD), density functional theory (DFT), and spectroscopic (Raman and IR) techniques. Atom−atom radial distribution functions (RDFs) based on MD simulation reveal that hydrogen bonds are formed between $Cl^-\cdots HO_{Ch^+}$ and $Cl^-\cdots HO_{COOH}$ of the THDES, where Cl^- works as a bridge between ASA and Ch^+. Cation−anion electrostatic attractions are disrupted by highly interconnected hydrogen bonds. Cluster conformers of the THDES are isolated from MD simulation and optimized using ωB97XD/6-311++G(d,p) level of theory, in which the strongest H bonds are found among $OH_{Ch^+}\cdots Cl^-$ (2.37 Å) and $Cl^-\cdots HO_{COOH}$(2.40 Å). Charge transfer calculations, using CHEPLG and NBO analysis, disclose that the charge of Cl^- is reduced in the cluster structures and transferred to Ch^+ and ASA. Further analyses are conducted using experimental and computed spectroscopic data. These confirm the formation of the THDES as peaks for −COOH, −COOR, and −OH functional groups of ASA and ChCl are either get broadened or disappeared in the spectra of the cluster conformers. Moreover, principal component analysis (PCA) assists to understand the feature of the simulated data and confirms the formation of the THDES. Solvent selectivity triangle (SST) of solvatochromic parameters also demonstrate that this THDES has some important properties similar to ionic liquids and common deep eutectic solvent.



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INTRODUCTION

Drug delivery faces substantial challenges as many newly developed drug molecules have low solubility, bioavailability, and stability.[1] Developing a new drug molecule is enormously expensive. Therefore, various techniques such as amorphous solid dispersion, lyophilization, particle size reduction, complexation, salt formation, and solvent formation have emerged to improve solubility and bioavailability of the existing active pharmaceutical ingredients (APIs).[2,3] It is easily understandable that improving the bioavailability of already existing drugs is way more profitable than launching a new one.[4]

Ionic liquids (ILs) have received great attraction in recent years to develop a potential pathway applying its features to overcome some of the problems of drug delivery mentioned above.[3] ILs have also been reported as drug delivery vehicles, such as formulation of microemulsions, improvement of dissolution, and enhancement of transdermal delivery of sparingly soluble drugs.[5,6] However, due to the expensive production, high toxicity, and poor biodegradability, drug delivery by ILs has become a less attractive option over the time.[7] A deep eutectic solvent (DES) is considered an analogue of ILs and a potential alternative solvent with a wide range of applications with less drawbacks.[8]

In DES, a hydrogen bond acceptor (HBA) and a hydrogen bond donor (HBD) are mixed together, where significant depression of freezing point is one of the most common phenomena.[9] Among different kinds of DESs, a therapeutic deep eutectic solvent (THDES) is a new type in which at least one of its components must be an API. Researchers successfully developed some THDESs to enhance their solubility, membrane transport, drug delivery, and bioavailability.[10,11] Formation of THDESs based on choline chloride, and menthol, combined with three different APIs including acetylsalicylic acid, benzoic acid, and phenylacetic acid were reported by Aroso and co-workers.[11] A few works have been reported on the basis of experimental studies and MD

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simulation of DES. Perkins and co-workers worked on choline chloride−urea eutectic solvent combining simulation and experiment.[12] Comparing MD simulations and infrared spectrum, they suggested that a strong interaction between the NH_2 of urea and the chloride anion occurs. They also studied three more choline chloride based eutectic solvents using the same methods and observed hydrogen bonding among the anions and HBDs, which agreed with their previous claim.[13] Sun et al. studied 1:2 ChCl/urea (reline) using classical MD simulation and concluded that the spontaneous intercalation of urea molecules in the choline chloride crystal lattice disrupted the ordered structure.[14] The simulated structure of the DES is in agreement with reported neutron diffraction structure.[15] Liquid reline and its mixture with water have also been probed by Fetisov et al.[16] and Shah et al.[17] by employing different force fields. Although several experimental studies are available on the THDES, little to no work has been reported on their structural, thermodynamic, and intermolecular interactions of the THDES with computational and experimental insights.

In this work, molecular dynamics (MD) simulation and density functional theory (DFT) calculations are employed to unravel the atomic level interactions of choline chloride (ChCl) based THDESs. Combined MD and DFT studies can assist to explain arrangements, orientations, interactions, and vibrations in the THDES.[18−21] The aims of this study are to find answers of the following questions: (i) what are the orientations of ChCl and acetylsalicylic acid (ASA) while they form the THDES, (ii) how is the melting point significantly decreased after forming the THDES, and (iii) what are the driving forces? Both computational and experimental Raman and FT-IR spectroscopic techniques[19] are employed to observe the molecular vibrations and the formation of hydrogen bonds in the THDES. Principle component analysis (PCA) is used to explain the spectra of different cluster conformers of the studied THDES further.[22] Solvatochromism analysis, which explains transition energy (E_T^N) and Kamlet−Taft parameters (hydrogen bond donor acidity (α), hydrogen bond acceptor basicity (β), and polarizability (π^*)[23]) is incorporated as a complementary method to explain how the studied THDES is different from or similar to other DESs, ILs, and common organic solvents.

METHODS

Computational Details. Molecular dynamics (MD) simulation using the YASARA Dynamics[24] program was used to generate all possible cluster conformers of the THDES. Here, the Amber 14[25] force field was considered and the particle mesh Ewald (PME) method[26] was used for calculating the long-range electrostatic interactions. Five, ten, and twenty pairs of ChCl and ASA were simulated in the gas phase for 10 ns. For initial comparison, the atom−atom radial distribution for one nonboning interaction was calculated for all three systems. In all cases, the peak was found at the same distance with different probabilities, which is obvious as the number of pairs are different (Figure S1). Among them, five pairs could not demonstrate the broad scenario of the solvent system and 20 pairs seemed too complex. Moreover, for big clusters, other parameters are difficult to calculate. In 10 pairs, simulation represented all types of interactions clearly (Figure S2); therefore, ten pairs were selected as the representative sample of the DES system for further simulation. The system was simulated for 40 ns to generate the atom−atom radial distribution function (RDF) of the system. The periodic boundary condition (cell box of 36.75 Å × 36.75 Å × 36.75 Å) and temperature of 298 K were considered using the *NVT* ensemble for all simulations. The system was minimized and equilibrated with the default protocol of YASARA dynamics. The time step of 1.25 fs was used and simulation snapshots were captured at every 100 ps. All simulations were repeated five times to reproduce the MD simulation results (Figure S3).

All quantum calculations were performed with the Gaussian 09 software package.[27] Gas phase equilibrium geometries of ChCl, ASA, and their most stable cluster conformers (generated from MD simulation) were fully optimized using the density functional theory (DFT) method, where the ωB97XD functional and 6-311++G(d,p) basis set were used to calculate the association energies of the clusters.[28] Changes of electronic energies, enthalpies, and the Gibbs free energies were also investigated to demonstrate the formation of strong nonbonding interactions between ChCl and ASA. Population analysis was carried out using the NBO method,[29] and the electrostatic potential derived charges were calculated employing the CHELPG scheme.[30]

Experimental Details. *Materials.* Nile red, 4-nitroaniline, *N*,*N*-diethyl-4-nitroaniline, and dichloromethane (DCM) were purchased from Thermo-Fischer Scientific (Dubuque, IA) as the UV−vis absorbance probes. Acetylsalicylic acid (ASA) and anhydrous choline chloride (ChCl) were purchased from Acros Organics, where all of the chemicals have purity ≥99% and analytical grade.

Synthesis of 1:1 ChCl/ASA THDES. ChCl/ASA (1:1) was formulated by mixing anhydrous choline chloride and acetylsalicylic acid in a 1:1 molar ratio at 80 °C for 3 h in a covered beaker under constant stirring 600 rpm at atmospheric pressure until a homogeneous liquid was obtained. After preparation, it was preserved in the desiccator because the compounds are highly hygroscopic.

FT-IR and Raman Spectroscopic Methods. A spectrum two PerkinElmer FT-IR spectrometer coupled with a $LiTaO_3$ (lithium tantalate) MIR detector was used to scan the 1:1 ChCl/ASA THDES sample. Spectra were recorded in transmission mode for an average of 20 individual scans in the spectral region 400−4000 cm^{-1} with a resolution of 1 cm^{-1} at room temperature. Background collection was done before every sampling, and data analysis was carried out using Origin Pro software version 9.0. The experiment was performed to analyze the presence of nonbonding intermolecular interactions between ChCl and ASA. For the analysis of solid ChCl and ASA, a Thermal Nicolet FT-IR 6700 was used with the same spectral region and resolution.

To measure the sensitivity of Raman spectrum of ChCl, ASA, and 1:1 ChCl/ASA, the HORIBA JobinYvon Raman (model-LabRAM 800HR) System was used, where spectral analysis was carried out by Origin Pro 9.0 software. The laser source (Green laser-MPC 6000) was powered on and the software linked to it was run in the computer. Samples were loaded at the sample holder and a 10× magnifying lens was focused on the samples, where the 530 nm wavelength laser (Green) was hitting over. The diffraction grating (300) and filter type were set via software to get the better scattered light for detection. In the case of the filter (D2), the lower the index was, the higher the intensity of scattered light would have been.

Principal Component Analysis (PCA) Method To Analyze Fingerprint Spectra. PCA is an exploratory statistical method that aims to reduce the dimensionality of the data to reduce

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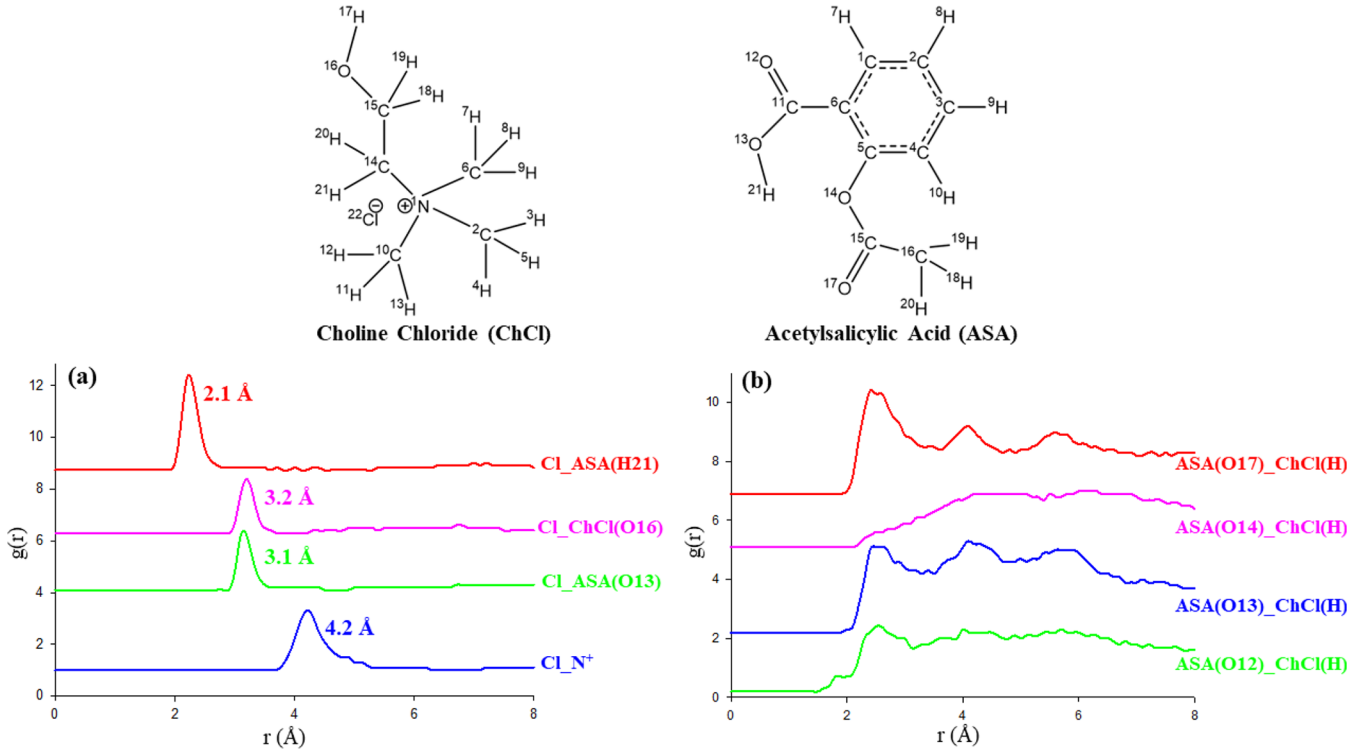


Figure 1. Atom−atom radial distribution functions (RDF) from MD simulations showing the probability of (a) Cl^- around different moieties and (b) hydrogen atoms around different oxygen atoms.

the noise as well as highlight the variabilities present in multivariate spectra of different chemical compounds.[31] PCA decomposes the multivariate response arranged in an **X** matrix into a product of two new matrices, as indicated by this equation: $\mathbf{X} = \mathbf{T}_k\mathbf{P}_k^T + \mathbf{E}$, where T_k is the matrix of scores that represent how samples relate to each other, $\mathbf{P}_k$ is the matrix of loadings that contain information about how variables relate to each other, k is the number of factors included in the model, and **E** is the matrix of residuals that contains the information not retained by the model. All calculations were performed with MATLAB 2015a (The Mathworks, Natick, MA, USA) equipped with the PLS Toolbox v. 6.2.1 (Eigenvector Research Inc., Wenatchee, WA, USA).[32]

Solvatochromic Method. Individual solutions of 5×10^{-4} M Nile red, 4-nitroaniline, and *N*,*N*-diethyl-4-nitroaniline dyes were prepared in dichloromethane (DCM). An appropriate amount of the dye stock solutions was micropipetted into clean, dry glass vials to mix with 1:1 ChCl/ASA. The glass vials were then capped, vortexed, and mixed properly. A 200 μL aliquot of each sample was loaded into a microplate. A UV−vis spectrophotometer (Synergy H1 microplate reader, Biotek) was used to determine the maximum absorption wavelength (λ_{max}) from the absorption spectra. Absorbance readings were collected as triplicate by spectral scanning of the THDES in Nile red (400−650 nm), *N*,*N*-diethyl-4-nitroaniline (300−450 nm), and 4-nitroaniline (300−450 nm) at 1 nm intervals.

RESULTS AND DISCUSSION

Radial Distribution Function (RDF) Analysis by MD Simulation. To elicit the formation of eutectic solvents and their nonbonding interactions, the atom−atom radial distribution functions (RDFs) are calculated from MD simulations. RDF can be used to investigate the distances between two atoms in MD simulations, which gives the probability of finding a particle at a certain distance from the reference particle compared to the statistical distribution.[15] The atom−atom RDFs, which provided the atomic level insights into the overarching structure of the THDES system, are represented in Figure 1. The complex nature of ChCl and ASA with multiple H bonding or nonbonding sites produces broad peaks and poorly defined minima. Figure 1a illustrates the contact distances between the atoms of ChCl and ASA. The first peak is between chloride and the hydrogen (21) atom on −OH of ASA, which shows the contact distances between the hydroxyl hydrogen atom and chlorine anion. This tall and narrow peak demonstrates the presence of hydrogen bonding and the interaction is around 2.1 Å, which is consistent with the HBD interactions with chloride anion in other previously studied DES systems.[12,13,33] It is known that the hydrogen bond strength can be correlated to the distance between hydrogen bond donor (HBD) and hydrogen bond acceptor (HBA).[34] Figure 1a also shows the RDF of the chloride and the HBD atom (O13) of the ASA as well as the RDF of Cl^- and O_{Ch^+} atom of choline cation. According to the RDFs, the distances between $Cl^-\cdots O_{Ch^+}$ and $Cl^-\cdots O_{ASA}$ are almost same, which signifies that Cl^- is working as a bridge between ASA and Ch^+. These phenomena are also confirmed from the simulation snapshots, depicted in Figure 2.

These simulations suggest that the disruption to cation−anion electrostatic attractions of ChCl by ASA and formation of highly interconnected hydrogen bonding in 1:1 ChCl/ASA could be one of the major rationales behind melting point depression. This disruption was estimated through the calculation of the N^+—Cl^- radial distribution function, shown in Figure 1a. Two peaks are observed in this RDF, a strong peak at 4.2 Å and a weak interaction at 4.8 Å. A previous computational study on choline chloride−urea, choline chloride−ethylene glycol, and choline chloride−glycerol

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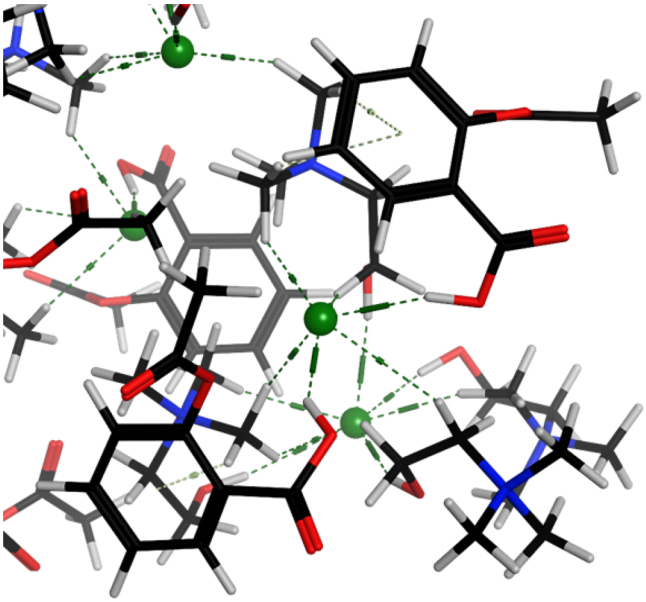


Figure 2. Simulation snapshots of the THDES environment showing chloride (green) as a bridge between ASA and Ch^+. Hydrogen atoms are in white, carbon atoms are in ash, nitrogen atoms are in blue, oxygen atoms are in red, and chlorides are in green.

found a similar feature, where the N^+—Cl^- radial distribution function was detected at ∼4.10 Å.[33] The strong peak of the RDF appearing at 4.2 Å can be attributed to the primary interaction (4.02 Å) in bulk choline chloride.[35] The weak interaction at 4.8 Å from the RDF can be compared with the secondary interactions (4.22 and 4.28 Å) in bulk choline chloride. It can be inferred that the distance between N^+—Cl^- according to RDF analysis is small enough that ASA cannot reside between the cation and anion. Figure 1b shows three strong nonboning intermolecular interactions among hydrogens (H) of ChCl and oxygen atoms from ASA, including esteric oxygen (O17) of the C═O group, carboxylic oxygen (O13) of the −OH group, and carboxylic oxygen (O12) of the C═O group. These peaks have distances less than 2.6 Å, which are most likely from hydrogen bonding.

These results also indicate that the nonbonding interactions involved in ChCl and ASA play the major role for the formation of the THDES at ambient temperature. All types of nonbonding interactions are labeled in a three-dimensional structure (Figure 3). For more clarification, a two-dimensional structure is also provided in Scheme S1. Among all of the interactions, hydrogen bonding between the chloride anion and the hydroxyl hydrogen (21) of ASA seems most important. Experimental IR data (given in the vibrational analysis section) of ChCl, ASA, and 1:1 ChCl/ASA also support the results obtained from RDF calculations.

Geometry and Hydrogen Bonding Analysis by DFT Calculation. The initial structures of 1:1 ChCl/ASA were isolated from MD simulation for the quantum calculation. Among those, the four most stable cluster conformers were considered for further optimization and calculation of their different quantum-chemical parameters. The pictorial illustrations of the four optimized conformers employing the ωB97XD/6-311++G(d, p) level of theory are depicted in Figure 4.

Structural orientations of a DES are important to understand its eutectic nature because the way molecules interact

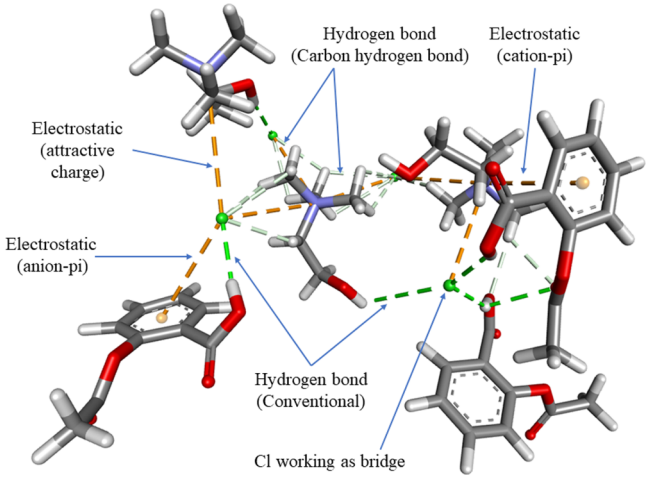


Figure 3. Simulation snapshots of the THDES environment showing all types of interactions present in the cluster systems. Hydrogen atoms are in white, carbon atoms are in ash, nitrogen atoms are in blue, and chloride ions are in green.

determines its physical properties, such as melting point, thermal stability, and vibrational spectra. Table S1 provides a list of hydrogen bond distances present in the cluster conformers and those are also labeled in Figure 4. The THDES system exhibits a highly interconnected hydrogen bonding network, wherein the ASA molecule interacts strongly with the Cl^- as well as the choline cation. Besides, the hydroxyl group of the carboxylic functionality on ASA interacts through the H···Cl^- H bond in all four conformers ranging from 1.98 to 2.10 Å. Analyzing F and Q conformers, it is found that the C═O end of the esteric functionality on ASA interacts with methyl protons on choline via C—H···π interactions with H···O distances of 2.35 and 2.29 Å, respectively. Moreover, in all four conformers, Cl^- is clearly acting as charge transfer bridge between the choline cation and ASA, which is further confirmed via charge transfer analysis.

Charge Transfer Analysis. According to a recent review on DESs, charge delocalization between the anion and HBDs is occurring via hydrogen bonding interactions, which is actually responsible for the depression of freezing point of DESs.[36] Again, the strength of natural and ionic hydrogen bond corelates with the amount of charge transfer (NBO) from the HBA to the HBD.[37]

The CHELPG partial charge analysis and NBO charge analysis were employed to investigate this proposed charge delocalization in our THDES system. Four different representative conformers were analyzed. As can be seen in Table 1, the charge of chloride is significantly reduced (ranging from −0.723 to −0.780) in all investigated conformers along with the net increase in the negative charge on the ASAs ranging from −0.042 to −0.075 according to CHELPG partial charge analysis. The same pattern of the results is observed in the case of NBO charge analysis as well. Comparing the charge distribution in the THDES system with its components reveals that the positive charge at the cationic core is increasing and the negative charge at the hydroxyl group of choline is decreasing. These analyses give clear evidence that both of the functional groups are essential for the charge transfer occurring among the choline cation, chloride anion, and acetylsalicylic acid molecule through hydrogen bonding and electrostatic interactions.

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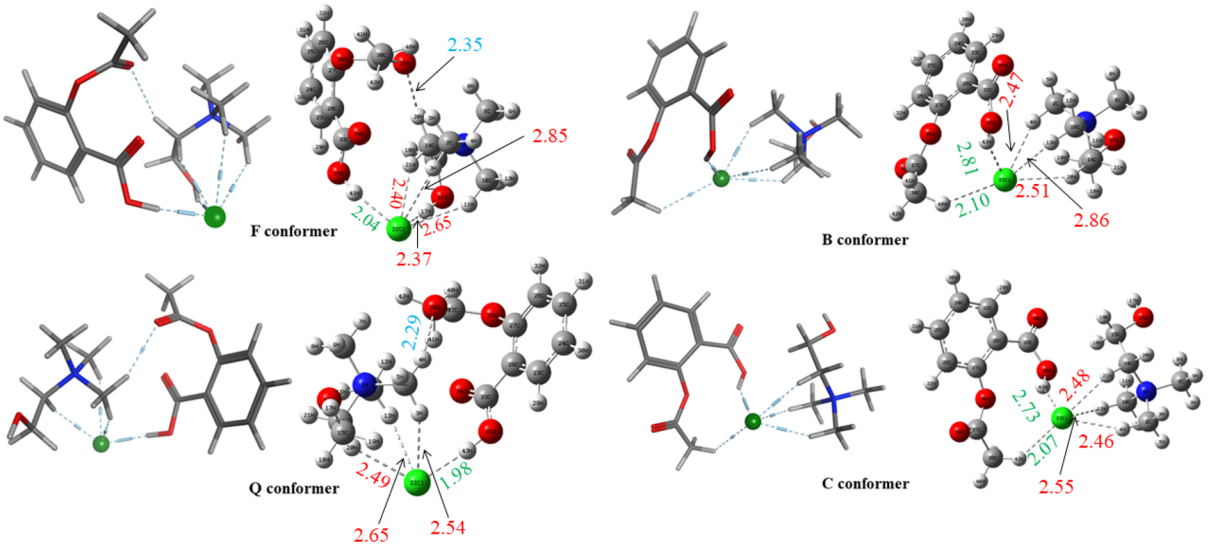


Figure 4. Optimized structures of four representative conformers (labeled as F, Q, B, and C) of THDES retrieved from MD simulation computed at the ωB97XD/6-311++G(d,p) level of theory. Hydrogen atoms are in white, carbon atoms are in ash, nitrogen atoms are in blue, and oxygen atoms are in red. The interatomic distances in red denote choline···Cl^-, in green denote ASA···Cl^-, and in blue denote choline···ASA interactions.

Table 1. CHELPG and NBO Charges on Different Components before and Formation of THDES Mixture for Four Different Conformers (Ch^+/Choline, Cl^-/Chloride, ASA/Acetylsalicylic Acid, Ch_N/N^+ of Choline) (Ch_O: Oxygen of Choline)

	CHELPG					NBO		
	Ch^+	Ch_N	Ch_O	Cl^-	ASA	Ch^+	Cl^-	ASA
F conformer	0.786	0.263	−0.678	−0.744	−0.042	0.91	−0.842	−0.064
Q conformer	0.789	0.293	−0.715	−0.743	−0.046	0.933	−0.846	−0.087
B conformer	0.788	0.256	−0.711	−0.743	−0.045	0.946	−0.871	−0.075
C conformer	0.798	0.171	−0.722	−0.723	−0.075	0.939	−0.857	−0.082
pure component	0.780	0.144	−0.729	−0.780	0	0.903	−0.903	0

Thermochemistry. In this study, the thermodynamic properties of ChCl, ASA, and the four cluster conformers of the studied THDES were calculated and the results are summarized in Table 2. Thermochemical values for each

Table 2. Change in Electronic Energies, Enthalpies, and Gibb's Free Energies Associated with Formation of Four Different Conformers Calculated at the DFT-ωB97XD/6-311++G(d,p) Level of Theory

combinations	Δ electronic energy (kJ/mol)	Δ enthalpy (kJ/mol)	Δ Gibb's free energy (kJ/mol)
B conformer	−75.38	−77.85	−28.75
C conformer	−70.76	−73.22	−13.94
F conformer	−98.61	−101.11	−47.23
Q conformer	−93.18	−95.65	−44.53

THDES conformers were obtained using the following equation: $\Delta X_{THDES} = X_{THDES} - X_{ChCl} - X_{ASA}$, where X is either electronic energy (E), enthalpy (H), or Gibbs free energy (G). A moderate change of electronic energy, enthalpy, and Gibb's free energy are observed. The results indicate that after forming the solvent system, it becomes more stabilized compared to their pure components in the gas phase. The corresponding electronic energy, enthalpy, and Gibb's free energy changes for the most stable conformer (F) are −98.61, −101.11, and −47.23 kJ/mol, respectively, and for the second most stable conformer (Q) are −93.18, −101.11, and −44.53 kJ/mol, respectively. This happened due to the presence of a higher number of nonbonding interactions in the F conformer (Figure 3). Analyses of all the studied THDES conformers show that there are a number of stable positions possible for ASA around ChCl.

Experimental and Calculated Raman and FTIR Spectra Analysis. Acetylsalicylic acid is an acetyl derivative of salicylic acid. Therefore, this compound should have two peaks between 1650 and 1850 cm^{-1} on its IR spectra for its acid and ester group.[38] In the 1:1 ChCl/ASA spectrum, two peaks in that range have been observed, where 1673 cm^{-1} corresponds to carboxylic acid group and 1751 cm^{-1} represents the ester group. However, both of the peaks are broadened and merged together due to the nonbonding intermolecular interactions between the ChCl and ASA. However, the peak for the −OH group of ChCl[12] was also broadened between 3135 and 3690 cm^{-1} in the spectrum of 1:1 ChCl/ASA, indicating strong H bonds between the −OH of ChCl and ASA (Figure 5b). Besides, in the range 900−980 cm^{-1}, the quaternary ammonium group[39] should show a characteristic peak. Here, a peak at 961 cm^{-1} represents the quaternary ammonium group of ChCl and another peak at 1232 cm^{-1} represents the C−N stretching vibration.[39] Peak broadening also occurs this time due to the strong doubly ionic effect.[20]

When looking at the experimental Raman spectra in Figure 5a, the most recognizable peak of the ASA spectrum was the C−C stretching vibrations at 1606 cm^{-1}, while several other prominent peaks are also observed in the ASA spectrum below 1400 cm^{-1}, which are 1294 and 1044 cm^{-1}. At around 3000

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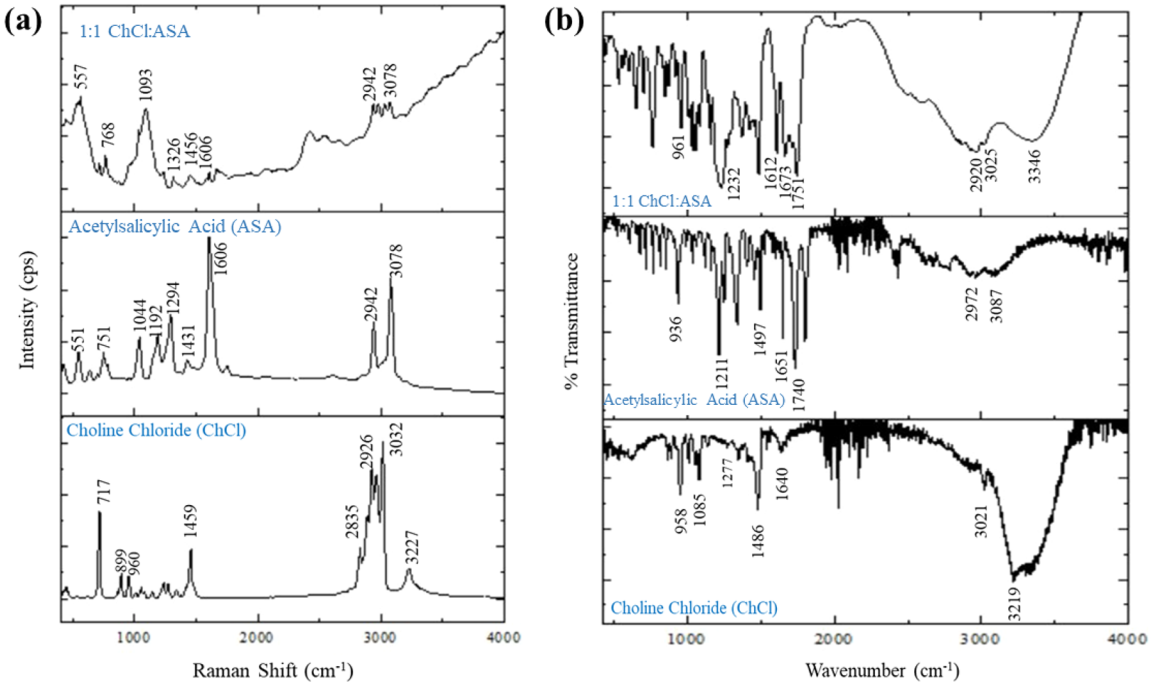


Figure 5. . (a) Raman spectra and (b) FTIR spectra of ChCl (bottom), ASA (middle) and THDES (top).

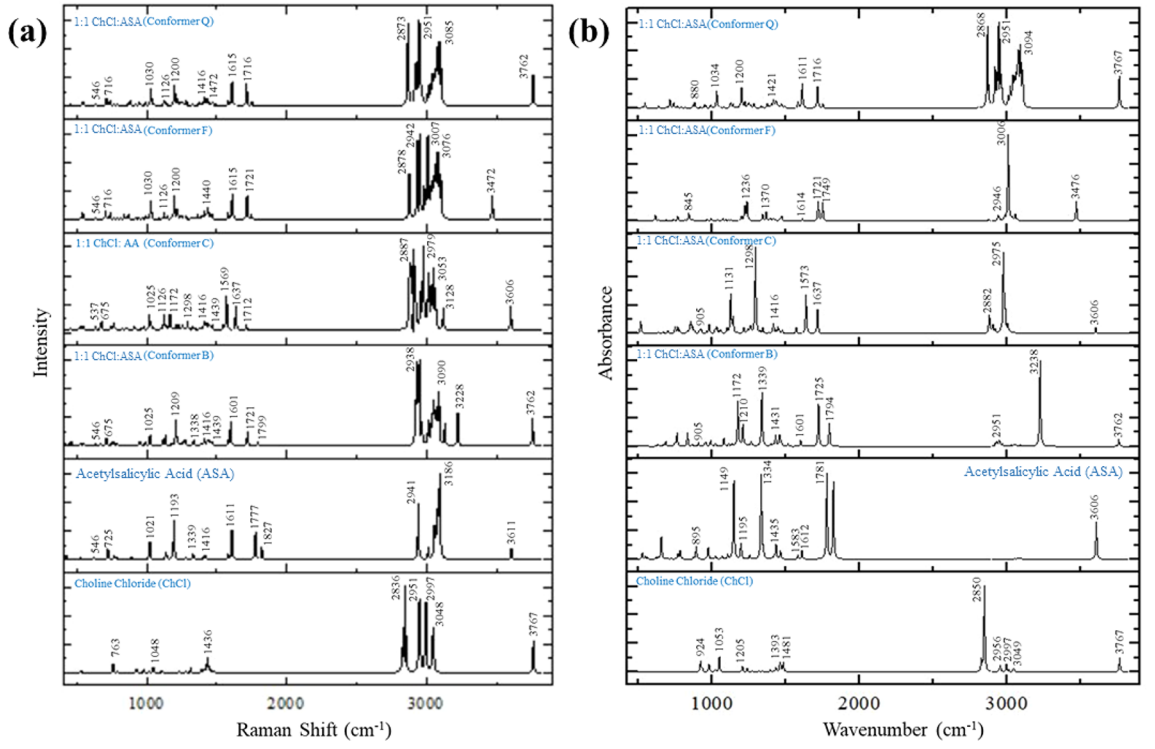


Figure 6. Calculated (a) Raman spectra and (b) infrared spectra of ChCl, ASA, and four different conformers (labeled as F, Q, B, and C).

cm^{-1}, two CH-stretching vibrations, 3078 and 2942 cm^{-1}, on different carbons were observed for ASA[40] whereas two major CH-stretching vibrations, 2926 and 3032 cm^{-1}, were observed on the ChCl spectrum. The symmetric νC−N peak at 717 cm^{-1} is intense in the ChCl spectrum.[41] All of the peaks of the ChCl and ASA spectra seem to nearly disappear or merge together in the 1:1 ChCl/ASA spectrum and appear as broad peaks. The disappearance or merger of the peaks is another indication for the formation of the THDES.[42]

To compare the experimental and the computed Raman and IR spectra, a scaling factor of 0.957[43] was used, as suggested in the NIST database. The calculated IR spectrum in Figure 6b shows that the peaks around 1700 cm^{-1}, specifically 1611 and 1777 cm^{-1}, seem to indicate the presence of the carboxylic acid and ester functional groups of ASA, respectively. These are similar to the peaks observed in the experimental IR spectrum of ASA. Different conformers, however, seem to have slightly different wavenumbers for the same peaks. Conformer Q only

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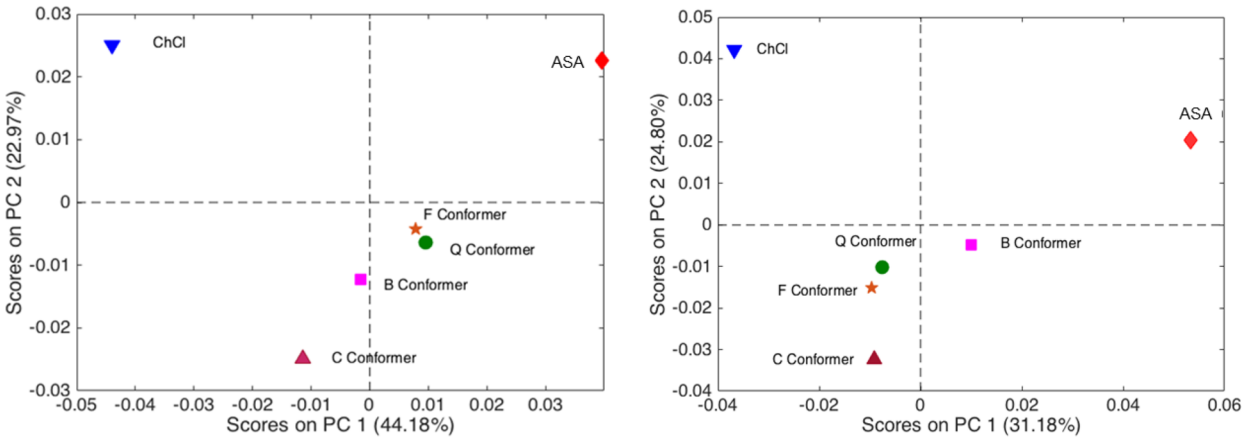


Figure 7. Score plots from principal components analysis of the (a) Raman and (b) IR simulated data (400−2000 cm^{-1}), specifically to compare complex fingerprint regions of the cluster conformers.

appears to have one peak attributable to esters, at 1716 cm^{-1}. Conformers F and B seem to have split peaks around this wavelength. Conformer F has peaks at 1721 and 1799 cm^{-1}, while conformer B has peaks at 1725 and 1794 cm^{-1}. Interestingly, the peaks for conformer B appear to be further shifted than the peaks for conformer F. ASA itself also seems to have a pair of peaks around 1700 cm^{-1}; however, they appear to be shifted toward larger wavenumbers, when the ASA is part of the THDES. This is a significant change to note. Peaks around 2900 cm^{-1} would most likely be attributed to the C−H stretching vibrations, which typically produce peaks in the 2800−3000 cm^{-1} range. These peaks are almost identical in their positioning in the spectra for conformer Q, conformer F, and conformer B. There are some of the peaks also seen in this range for the choline chloride. The majority of the calculated spectra also show a peak in the 3600−3800 cm^{-1} range, which is greatly separated from the rest of the peaks on the spectrum. This is highly likely due to the alcohol functional group of choline chloride. It is not confirmed why a peak in this range appeared for the acetylsalicylic acid. However, this is likely due to the −OH group on the carboxylic acid.

Again, Raman spectra in Figure 6a show a couple of very distinct peaks, but similar to experimental peaks. There is a well-defined C−C stretching vibrations found between 1600 and 1640 cm^{-1} in the spectra of all four of the conformers and ASA. Similar to the experimental peaks of ASA, below 1400 cm^{-1}, two very definitive peaks at 1021 and 1193 cm^{-1} indicate the conformity of the calculated spectrum with the experimental spectrum. All of the spectra also show peaks around 2800−3100 cm^{-1}, which are the symmetric C−H vibrations similar to the experimental Raman peaks in the same region (Figure 6a). It has also been observed that peaks of ChCl and ASA in this range merged together and emerge as a higher number of peaks, which indicate the formation of the THDES.

It is difficult to distinguish the fingerprint region peaks of the spectra of four conformers. Therefore, in an attempt to identify the significant differences of four conformers in that region along with ChCl and ASA, simulated spectra were analyzed using a principal component analysis (PCA) algorithm. Spectral data are truncated to 400−2000 cm^{-1} for the convenience of the analysis as zero intensities are present in the longer wavelengths. Spectra are preprocessed using normalization and mean centering to highlight the separation of different groups. Data are cross-validated using the leave-one-out method while doing the analysis.

Figure 7 (panels a and b) illustrates the distribution of the pure components (ChCl and ASA) and four conformers in the PC1 and PC2 score axis. It is observed that PC1 can successfully separate scores of ChCl and ASA for both Raman and IR data. It is rational as their spectral features are distinct from each other. Scores of the four conformers reside between those of ChCl and ASA, which confirms the spectral features of these four conformers are the combination of ChCl and ASA. However, the score generated from the IR spectrum of the B conformer is located near that for ASA in the PC1 score axis (Figure 7b). This is probably due to the analogous location (around 1700 cm^{-1}) of the ester group of these two components, i.e., ASA and conformer B. On the contrary, scores of all four conformers, generated from Raman spectra, reside near the ASA score (Figure 7a). As Raman spectroscopy is sensitive toward the C−C stretching vibration, peaks appear between 1600 and 1640 cm^{-1} in the four conformers and ASA, which has probably positioned all the conformers near ASA. Therefore, PCA analysis assists us to visualize the similarity and dissimilarity of IR and Raman spectral data of ChCl, ASA, and THDES conformers.

Solvatochromic Parameter Analysis. Although all of the solvatochromic parameters are obtained from 25 to 45 ± 0.2 °C at 5 °C intervals, our focus is mostly on the parameters obtained at ambient temperature, as increasing temperature does not show any significant effect on the properties. At ambient temperature, the transition energy, E_T(NR) and E_T^N(dimensionless), values are calculated from the solvatochromic responses of the UV−vis probes and are found to be 47.49 and 0.518, respectively. Hydrogen bond donor (HBD) acidity (α) and hydrogen bond acceptor (HBA) basicity (β) show the measurement of the potential of a HBD to donate a proton to a HBA, and the potential of a HBA to receive a proton from a HBD, respectively. At 25 °C, the α value calculated for this THDES is found to be 0.259, and the β value is 0.553, which suggest that Cl^- is more prone to take a proton from the system. The polarity of the solvent measures the ability of the solvent to hold charge due to dielectric effect.[44] The solvent polarity (π^*) should decrease with increasing temperature due to the thermal reorientation of

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dipoles, which ultimately decreases the dielectric constant.[45] The polarity of the studied THDES has followed this theoretical basis as, at 25 °C, the π^* value is 1.1023; however, at 45 °C, it decreases to 1.0923.

To get a comparative visualization of the Kamlet−Taft parameters (α, β, and π^*) of the studied THDES with some ILs, organic solvents, and DESs, a ternary plot (Figure 8) is

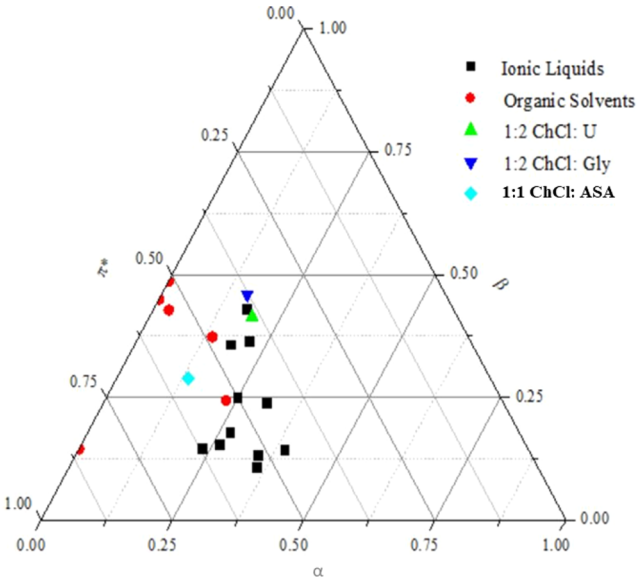


Figure 8. Kamlet−Taft parameters of 1:1 ChCl/ASA at 298 K compared to the selected organic solvents, ionic liquids, and the most common DESs.

constructed by Origin Pro 9.0 software packageusing the data[46−52] and is tabulated in Table S2. Snyder introduced this type of solvent selectivity triangle (SST) to characterize the solvents and group them by plotting α, β, and π^* of the solvents.[53] The ternary plot in Figure 8 indicates that the Kamlet−Taft parameters of 1:1 ChCl/ASA are closer to the parameters of the ionic liquids than to those of the organic solvents. Therefore, it is clearly a better alternative to ILs as it shares most of the excellent properties of ILs, but is less toxic, easy to formulate, and less expensive.

CONCLUSION

In this study, we have investigated the interactions between ChCl and ASA in the THDES by combined computational and experimental techniques. Analyzing RDF data generated from MD simulations demonstrates that there are several hydrogen bonds between ChCl and ASA, especially between the −OH of ASA and Cl^- of ChCl. Density functional theory calculations show that Cl^- is acting as a charge transfer bridge between Ch^+ and ASA, which is observed in RDF analysis. The CHELPG partial charge analysis and NBO charge analysis reveal that the positive charge at the cationic core and the hydroxyl group of Ch^+ have been decreased and transferred to ASA through the chloride bridge. Kamlet−Taft parameters, α, β, and π^*, experimentally confirmed that Cl^- plays the most important role to transfer charges in the hydrogen bonding network of the THDES. Thermochemical data acquired from the DFT calculation proved the spontaneity of the formation of its conformers.

Experimental Raman and IR spectra of 1:1 ChCl/ASA are compared with the spectra of ChCl and ASA. Some of the peaks of the ChCl and ASA spectra are nearly disappeared or are merged in the spectrum of 1:1 ChCl/ASA, which is the clear indication of hydrogen bond formation or the formation of the THDES. The computed Raman and IR spectra showed similar features; however, computed spectra show more peaks compared to the experimental spectra. Hence, it is difficult to differentiate the peaks located below 2000 cm^{-1} or specifically the peaks of the fingerprint region. Principal component analysis not only effectively assists to compare the spectra of ChCl, ASA, and the four cluster conformers of 1:1 ChCl/ASA of that region but also demonstrates that the spectral features of ChCl and ASA are totally distinctive and the conformer's spectral features are between those of ChCl and ASA. The results obtained from this study can be summarized in three points: (i) MD simulation generates the representative conformers, which show the possible orientations of the microstructures of ChCl/ASA. (ii) Radial distribution function (RDF) and spectral data analyses confirm that the hydrogen bonding and the interaction patterns involved in the system cause the depression of the melting point. (iii) DFT calculations and solvatochromic parameters provide the information about charge dislocation, which is the major driving force for the formation of the THDES. In summary, this work shows in-depth atomic level understanding of how a THDES is formed.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpca.0c00851.

RDFs, simulation snapshots for 5, 10, and 20 pairs of ChCl and ASA in the DES system, energy profile, two-dimensional structure of the solvent system to show the interactions, a table for the selected bond distances of the four conformers calculated with DFT, and a table for the preparation of the ternary plot using the listed solvatochromic parameters of selected solvents (PDF)

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Notes

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■ REFERENCES

(1) Kalepu, S.; Nekkanti, V. Insoluble Drug Delivery Strategies: Review of Recent Advances and Business Prospects. *Acta Pharm. Sin. B* **2015**, *5*, 442−453.
(2) Savjani, K. T.; Gajjar, A. K.; Savjani, J. K. Drug Solubility: Importance and Enhancement Techniques. *ISRN Pharm.* **2012**, *2012*, 1−10.
(3) Hough, W. L.; Smiglak, M.; Rodríguez, H.; Swatloski, R. P.; Spear, S. K.; Daly, D. T.; Pernak, J.; Grisel, J. E.; Carliss, R. D.; Soutullo, M. D.; et al. The Third Evolution of Ionic Liquids: Active Pharmaceutical Ingredients. *New J. Chem.* **2007**, *31*, 1429.
(4) Saadi, E.; White, G. Rewarding Innovation in Drug Development. *Am. Heal. Drug Benefits* **2014**, *7*, 373−374.
(5) Moniruzzaman, M.; Tamura, M.; Tahara, Y.; Kamiya, N.; Goto, M. Ionic Liquid-in-Oil Microemulsion as a Potential Carrier of Sparingly Soluble Drug: Characterization and Cytotoxicity Evaluation. *Int. J. Pharm.* **2010**, *400*, 243−250.
(6) Moniruzzaman, M.; Kamiya, N.; Goto, M. Ionic Liquid Based Microemulsion with Pharmaceutically Accepted Components: Formulation and Potential Applications. *J. Colloid Interface Sci.* **2010**, *352*, 136−142.
(7) Paiva, A.; Craveiro, R.; Aroso, I.; Martins, M.; Reis, R. L.; Duarte, A. R. C. Natural Deep Eutectic Solvents − Solvents for the 21st Century. *ACS Sustainable Chem. Eng.* **2014**, *2*, 1063−1071.
(8) Abbott, A. P.; Boothby, D.; Capper, G.; Davies, D. L.; Rasheed, R. K. Deep Eutectic Solvents Formed between Choline Chloride and Carboxylic Acids: Versatile Alternatives to Ionic Liquids. *J. Am. Chem. Soc.* **2004**, *126*, 9142−9147.
(9) Smith, E. L.; Abbott, A. P.; Ryder, K. S. Deep Eutectic Solvents (DESs) and Their Applications. *Chem. Rev.* **2014**, *114*, 11060−11082.
(10) Stott, P. W.; Williams, A. C.; Barry, B. W. Transdermal Delivery from Eutectic Systems: Enhanced Permeation of a Model Drug, Ibuprofen. *J. Controlled Release* **1998**, *50*, 297−308.
(11) Aroso, I. M.; Silva, J. C.; Mano, F.; Ferreira, A. S. D.; Dionísio, M.; Sá-Nogueira, I.; Barreiros, S.; Reis, R. L.; Paiva, A.; Duarte, A. R. C. Dissolution Enhancement of Active Pharmaceutical Ingredients by Therapeutic Deep Eutectic Systems. *Eur. J. Pharm. Biopharm.* **2016**, *98*, 57−66.
(12) Perkins, S. L.; Painter, P.; Colina, C. M. Molecular Dynamic Simulations and Vibrational Analysis of an Ionic Liquid Analogue. *J. Phys. Chem. B* **2013**, *117*, 10250−10260.
(13) Perkins, S. L.; Painter, P.; Colina, C. M. Experimental and Computational Studies of Choline Chloride-Based Deep Eutectic Solvents. *J. Chem. Eng. Data* **2014**, *59*, 3652−3662.
(14) Sun, H.; Li, Y.; Wu, X.; Li, G. Theoretical Study on the Structures and Properties of Mixtures of Urea and Choline Chloride. *J. Mol. Model.* **2013**, *19*, 2433−2441.
(15) Hammond, O. S.; Bowron, D. T.; Edler, K. J. Liquid Structure of the Choline Chloride-Urea Deep Eutectic Solvent (Reline) from Neutron Diffraction and Atomistic Modelling. *Green Chem.* **2016**, *18*, 2736−2744.
(16) Fetisov, E. O.; Harwood, D. B.; Kuo, I-F. W.; Warrag, S. E. E.; Kroon, M. C.; Peters, C. J.; Siepmann, J. I. First-Principles Molecular Dynamics Study of a Deep Eutectic Solvent: Choline Chloride/Urea and Its Mixture with Water. *J. Phys. Chem. B* **2018**, *122*, 1245−1254.
(17) Shah, D.; Mjalli, F. S. Effect of Water on the Thermo-Physical Properties of Reline: An Experimental and Molecular Simulation Based Approach. *Phys. Chem. Chem. Phys.* **2014**, *16*, 23900−23907.
(18) Allen, S. J.; Curran, A. R.; Templer, R. H.; Meijberg, W.; Booth, P. J. Controlling the Folding Efficiency of an Integral Membrane Protein. *J. Mol. Biol.* **2004**, *342*, 1293−1304.
(19) Wu, X.; Gao, S.; Wang, J.-S.; Wang, H.; Huang, Y.-W.; Zhao, Y. The Surface-Enhanced Raman Spectra of Aflatoxins: Spectral Analysis, Density Functional Theory Calculation, Detection and Differentiation. *Analyst* **2012**, *137*, 4226.
(20) Ashworth, C. R.; Matthews, R. P.; Welton, T.; Hunt, P. A. Doubly Ionic Hydrogen Bond Interactions within the Choline Chloride-Urea Deep Eutectic Solvent. *Phys. Chem. Chem. Phys.* **2016**, *18*, 18145−18160.
(21) Lindahl, E. R. Molecular Dynamics Simulations. *Methods Mol. Biol.* **2008**, *443*, 3−23.
(22) Wang, Q.; Gao, Q.; Gao, X.; Nie, F. Angle Principal Component Analysis. *IJCAI Int. Jt. Conf. Artif. Intell.* **2017**, *2*, 2936−2942.
(23) Taft, R. W.; Pienta, N. J.; Kamlet, M. J.; Arnett, E. M. Linear Solvation Energy Relationships. 7. Correlations between the Solvent-Donicity and Acceptor-Number Scales and the Solvatochromic Parameters. Pi. *,. Alpha. and Beta. *J. Org. Chem.* **1981**, *46*, 661−667.
(24) Krieger, E.; Vriend, G. New Ways to Boost Molecular Dynamics Simulations. *J. Comput. Chem.* **2015**, *36*, 996−1007.
(25) Maier, J. A.; Martinez, C.; Kasavajhala, K.; Wickstrom, L.; Hauser, K. E.; Simmerling, C. Ff14SB: Improving the Accuracy of Protein Side Chain and Backbone Parameters from Ff99SB. *J. Chem. Theory Comput.* **2015**, *11*, 3696−3713.
(26) Darden, T.; York, D.; Pedersen, L. Particle Mesh Ewald: An N·log(N) Method for Ewald Sums in Large Systems. *J. Chem. Phys.* **1993**, *98*, 10089−10092.
(27) Frisch, M. J.; T, G. W.; Schlegel, H. B.; Scuseria, G. E.; Robb, M. A.; Cheeseman, J. R.; Scalmani, G.; Barone, V.; Mennucci, B.; Petersson, G. A.; et al. *Gaussian 09*, Revision A. 02; Gaussian, Inc.: Wallingford, CT, 2009.
(28) Burns, L. A.; Mayagoitia, A. V.; Sumpter, B. G.; Sherrill, C. D. Density-Functional Approaches to Noncovalent Interactions: A Comparison of Dispersion Corrections (DFT-D), Exchange-Hole Dipole Moment (XDM) Theory, and Specialized Functionals. *J. Chem. Phys.* **2011**, *134*, 084107.
(29) Glendening, E. D.; Landis, C. R.; Weinhold, F. Natural Bond Orbital Methods. *Wiley Interdiscip. Rev.: Comput. Mol. Sci.* **2012**, *2*, 1−42.
(30) Breneman, C. M.; Wiberg, K. B. Determining Atom-centered Monopoles from Molecular Electrostatic Potentials. The Need for High Sampling Density in Formamide Conformational Analysis. *J. Comput. Chem.* **1990**, *11*, 361−373.
(31) Sanchez, J.; Martens, H.; Naes, T.. *Multivariate Calibration*; John Wiley & Sons: Chichester, 1989; 419+xvii pp. *Biometrical J.* **1991**, *33*, 418−418
(32) Igne, B.; Talwar, S.; Feng, H.; Drennen, J. K.; Anderson, C. A. Near-Infrared Spatially Resolved Spectroscopy for Tablet Quality Determination. *J. Pharm. Sci.* **2015**, *104*, 4074−4081.
(33) Stefanovic, R.; Ludwig, M.; Webber, G. B.; Atkin, R.; Page, A. J. Nanostructure, Hydrogen Bonding and Rheology in Choline Chloride Deep Eutectic Solvents as a Function of the Hydrogen Bond Donor. *Phys. Chem. Chem. Phys.* **2017**, *19*, 3297−3306.
(34) Wendler, K.; Thar, J.; Zahn, S.; Kirchner, B. Estimating the Hydrogen Bond Energy. *J. Phys. Chem. A* **2010**, *114*, 9529−9536.

(35) Senko, M. E.; Templeton, D. H. Unit Cells of Choline Halides and Structure of Choline Chloride. *Acta Crystallogr.* **1960**, *13*, 281−285.
(36) Carriazo, D.; Serrano, M. C.; Gutiérrez, M. C.; Ferrer, M. L.; del Monte, F. Deep-Eutectic Solvents Playing Multiple Roles in the Synthesis of Polymers and Related Materials. *Chem. Soc. Rev.* **2012**, *41*, 4996.
(37) Meot-Ner, M. The Ionic Hydrogen Bond. *Chem. Rev.* **2005**, *105* (1), 213−284.
(38) Binev, I. G.; Stamboliyska, B. A.; Binev, Y. I. The Infrared Spectra and Structure of Acetylsalicylic Acid (Aspirin) and Its Oxyanion: An Ab Initio Force Field Treatment. *J. Mol. Struct.* **1996**, *378*, 189−197.
(39) Mendoza, A. J. *Thermal and Electrochemical Degradation of Anion Exchange Membranes*. Ph.D. Dissertation, The Pennsylvania State University, PA, USA, 2012.
(40) Szostak, R.; Mazurek, S. Quantitative Determination of Acetylsalicylic Acid and Acetaminophen in Tablets by FT-Raman Spectroscopy. *Analyst* **2002**, *127*, 144−148.
(41) Araujo, C. F.; Coutinho, J. A. P.; Nolasco, M. M.; Parker, S. F.; Ribeiro-Claro, P. J. A.; Rudić, S.; Soares, B. I. G.; Vaz, P. D. Inelastic Neutron Scattering Study of Reline: Shedding Light on the Hydrogen Bonding Network of Deep Eutectic Solvents. *Phys. Chem. Chem. Phys.* **2017**, *19*, 17998−18009.
(42) Zhu, S.; Li, H.; Zhu, W.; Jiang, W.; Wang, C.; Wu, P.; Zhang, Q.; Li, H. Vibrational Analysis and Formation Mechanism of Typical Deep Eutectic Solvents: An Experimental and Theoretical Study. *J. Mol. Graphics Modell.* **2016**, *68*, 158−175.
(43) Ritcher, U.; Manichaikul, A.; Schneider, B.; Acevedo, C.; Cockrell, B.; Minne, E.; Reed, J.; Parsons, M.; Fahey, M. Precomputed vibrational scaling factors, Computational Chemistry Comparison and Benchmark DataBase; Johnson, R. D., Eds.; *NIST Standard Reference Database 101*; National Institute of Standards and Technology, Gaithersburg, MD, https://cccbdb.nist.gov/vibscalejust.asp (retrieved Aug 9, 2019).
(44) Kamlet, M. J.; Abboud, J. L.; Taft, R. W. The Solvatochromic Comparison Method. 6. The. Pi.* Scale of Solvent Polarities. *J. Am. Chem. Soc.* **1977**, *99*, 6027−6038.
(45) Marcus, Y. *Introduction to Liquid State Chemistry*; Wiley: New York, 1977.
(46) Wu, Y.; Sasaki, T.; Kazushi, K.; Seo, T.; Sakurai, K. Interactions between Spiropyrans and Room-Temperature Ionic Liquids: Photochromism and Solvatochromism. *J. Phys. Chem. B* **2008**, *112*, 7530−7536.
(47) Zhang, S.; Qi, X.; Ma, X.; Lu, L.; Deng, Y. Hydroxyl Ionic Liquids: The Differentiating Effect of Hydroxyl on Polarity Due to Ionic Hydrogen Bonds between Hydroxyl and Anions. *J. Phys. Chem. B* **2010**, *114*, 3912−3920.
(48) Mellein, B. R.; Aki, S. N. V. K.; Ladewski, R. L.; Brennecke, J. F. Solvatochromic Studies of Ionic Liquid/Organic Mixtures. *J. Phys. Chem. B* **2007**, *111*, 131−138.
(49) Khupse, N. D.; Kumar, A. Delineating Solute-Solvent Interactions in Binary Mixtures of Ionic Liquids in Molecular Solvents and Preferential Solvation Approach. *J. Phys. Chem. B* **2011**, *115*, 711−718.
(50) Chiappe, C.; Pieraccini, D. Determination of Ionic Liquids Solvent Properties Using an Unusual Probe: The Electron Donor-Acceptor Complex between 4,4'-Bis(Dimethylamino)-Benzophenone and Tetracyanoethene. *J. Phys. Chem. A* **2006**, *110*, 4937−4941.
(51) Crowhurst, L.; Mawdsley, P. R.; Perez-Arlandis, J. M.; Salter, P. A.; Welton, T. Solvent−Solute Interactions in Ionic Liquids. *Phys. Chem. Chem. Phys.* **2003**, *5*, 2790−2794.
(52) Degam, G. Deep Eutectic Solvents Synthesis, Characterization and Applications in Pretreatment of Lignocellulosic Biomass. *Ph.D. Dissertation*, South Dakota State University, Brookings, SD, 2017.
(53) Snyder, L. R. Classification off the Solvent Properties of Common Liquids. *J. Chromatogr. Sci.* **1978**, *16*, 223−234.

https://dx.doi.org/10.1021/acs.jpca.0c00851
J. Phys. Chem. A 2020, 124, 4690−4699